

Activity

1. Medical Image Enhancement

1. which gray-level transformation would you recommend?

Ans power-law (Gamma) transformation with $\gamma < 1$

2. why is it suitable?

Because it enhances the dark regions of the x-ray without significantly affecting the already bright bones. This improves the visibility of soft tissues.

3. what effect will it have on the image?

- * Dark tissues become brighter and clearer
- * Soft tissue details become more visible
- * Overall image contrast improves while bones remain visible

② Night-Time surveillance

1. which transformation would improve the visibility?

Ans log transformation.

2. why is it more suitable than negative transformation?

log transformation expands low intensity (dark) pixel values, making hidden details visible.

Negative transformation only inverts the image colors and does not effectively improve visibility in dark images.

3. predict the appearance of the enhanced image

* the image becomes brighter.

* cars, people, and objects are easier to identify.

* Details in dark areas become more visible.

③ Fingerprint Recognition

1. which gray-level transformation would you apply?

Ans Contrast stretching (piecewise linear transformation).

2. Explain why.

Contrast stretching increases the difference between fingerprint ridges and valleys, making the fine ridge details clearer for accurate recognition.

3. what would happen if a log transformation were applied instead?

log transformation would brighten the darker regions but would not effectively enhance the contrast b/w ridges and valleys. As a result, the fingerprint details may become less distinct, reducing recognition accuracy.